Chapter-01: Introduction to Automata Theory

- **What is automata theory?**
  Study of abstract computing devices, or "machines".
- **Automation:** An abstract computing device/A "device" need not even be a physical hardware!
- **Theory of computation:** The theory of computation studies *what different types of machines can and cannot do* in terms of problem-solving. It helps us understand the limits of computing.
  Example:
  - ✓ A calculator can quickly *add or multiply numbers* (simple tasks ✅).
  - ✓ But it cannot understand *natural language or play chess* (complex tasks ❌).

  Different models of computation, like **Finite Automata, Pushdown Automata, and Turing Machines**, help define these limits. A Turing Machine is the most powerful model, capable of solving any problem a computer can compute!
- **Computability vs. Complexity**

Feature	Computability	Complexity
Key Concern	Can we solve it?	How fast can we solve it?
Definition	Checks **if** a problem can be solved by a machine.	Measures **how efficiently** a problem can be solved.
Focus	**Possibility** of solving a problem.	**Resources** (time, memory) needed to solve a problem.
Types of Problems	**Decidable (solvable)** & **Undecidable (unsolvable)** problems.	**Easy (P), Hard (NP, NP-Hard, NP-Complete)** problems.
Example (Easy)	Checking if a number is even (✅ Yes, it's computable).	Sorting a list of numbers (Fast).
Example (Hard)	Halting Problem (❌ Cannot be solved by any algorithm).	Sudoku puzzle solving (Takes time).

- **Turing test:** The Turing Test, proposed by Alan Turing (1950), is a test to check if a machine (AI) can think like a human.

How It Works:

1. A **human judge** chats with both a **human** and a **machine (AI)** via text.
2. The judge **doesn't know** which one is human.
3. If the AI **fools the judge** into thinking it's human, it **passes** the test!

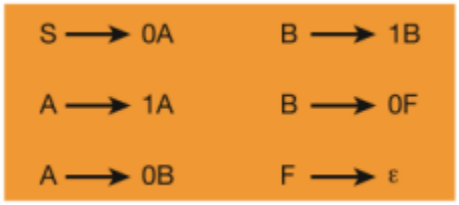
Example:

- **AI Chatbots** like **ChatGPT, Deepseek** or **Siri** try to have human-like conversations.
- If a chatbot responds naturally and the judge can't tell it's a machine, it **passes** the Turing Test.

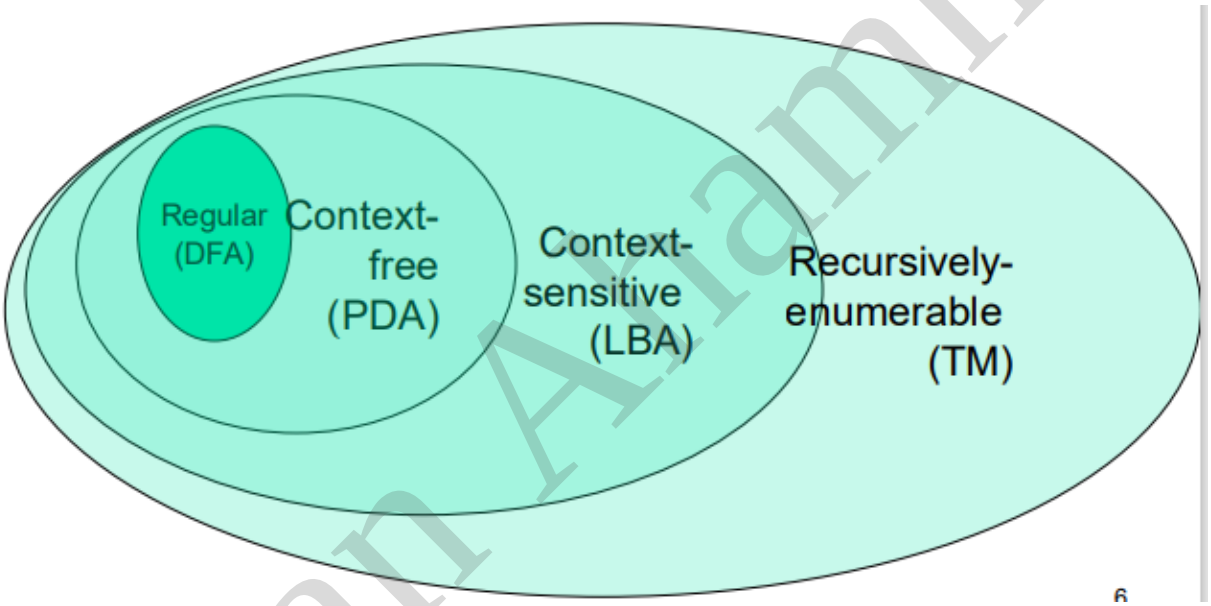
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❖ **Languages and Grammars:**

- **Alphabet:** An alphabet is a finite, non-empty set of symbols- {0,1}
- **String/Word:** A string or word is a finite sequence of symbols -{01…}
- **Sentences:** Sentences are string of symbols- {0, 1, 00, 01, 10, 1,….}
- **Language:** A language is a set of sentences– {000, 0100, 0010,……..}
- **Grammar:** A grammar is a finite list of rules defining a language:



❖ **The Chomsky Hierachy**



Type	Grammar	Automaton	Example
Type 0	Unrestricted (any rules allowed)	Turing Machine	All valid computer programs
Type 1	Context-sensitive (rules expand string)	Linear-Bounded Automaton	Balanced parentheses (()())
Type 2	Context-free (rules expand non-terminals)	Pushdown Automaton	Mathematical expressions 3 + (4 * 5)
Type 3	Regular (simple rules)	Finite Automaton	Binary strings with pattern 0*

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❖ **The central concepts of automata theory:**

1. **Alphabet:** An alphabet is a finite, non-empty set of symbols.
   - ➢ We use the symbol ∑ (sigma) to denote an alphabet.
   - ➢ Examples:
     - Binary: ∑ = {0,1}
     - All lower case letters: ∑ = {a,b,c,..z}
     - Alphanumeric: ∑ = {a-z, A-Z, 0-9}
     - DNA molecule letters: ∑ = {a,c,g,t}
2. **Strings:** A string or word is a finite sequence of symbols\
   - ➢ Empty string is ε (or "epsilon")
   - ➢ Length of a string w, denoted by "|w|", is equal to the number of (non- □) characters in the string.
     - E.g., x = 010100 |x| = 6
     - x = 01 ε 0 ε 1 ε 00 ε |x| = ?
     - xy = Concatentation of two strings x and y
3. **Power of an alphabet:** Let ∑ be an alphabet.
   - ➢ $\sum^0 = \{\varepsilon\}$, regardless of what alphabet ∑ is (ε is the only string whose length is 0).
   - ➢ if ∑ = {0, 1},
     - Then $\sum^1 = \{0, 1\}$
     - $\sum^2 = \{00, 01, 10, 11\}$
     - $\sum^3 = \{000, 001, 010, 011, 100, 101, 110, 111\}$
   - ➢ Let ∑ be an alphabet.
     - $\sum^k$ = the set of all strings of length k
     - $\sum^* = \sum^0 \cup \sum^1 \cup \sum^2 \cup \ldots$ (set of all strings over an alphabet)
     - $\sum^+ = \sum^1 \cup \sum^2 \cup \sum^3 \cup \ldots$ (set of nonempty strings)
4. **Languages:**
   - ➢ L is a said to be a language over alphabet ∑, only if $L \subseteq \sum^*$. This is because ∑* is the set of all strings (of all possible length including 0) over the given alphabet ∑.
   - ➢ **Examples:**
     1. Let L be the language of all strings consisting of n 0's followed by n 1's:
        L = {ε, 01, 0011, 000111,…}
     2. Let L be the language of all strings of with equal number of 0's and 1's:
        L = {ε, 01, 10, 0011, 1100, 0101, 1010, 1001,…}
   - ➢ Definition: Ø denotes the Empty language
   - ➢ Let L = {ε}; Is L=Ø? NO

❖ **Given a string w ε ∑*and a language L over ∑, decide whether or not w ε L.**

Example: Let w = 100011

➢ Is w the language of strings with equal number of 0s and 1s?

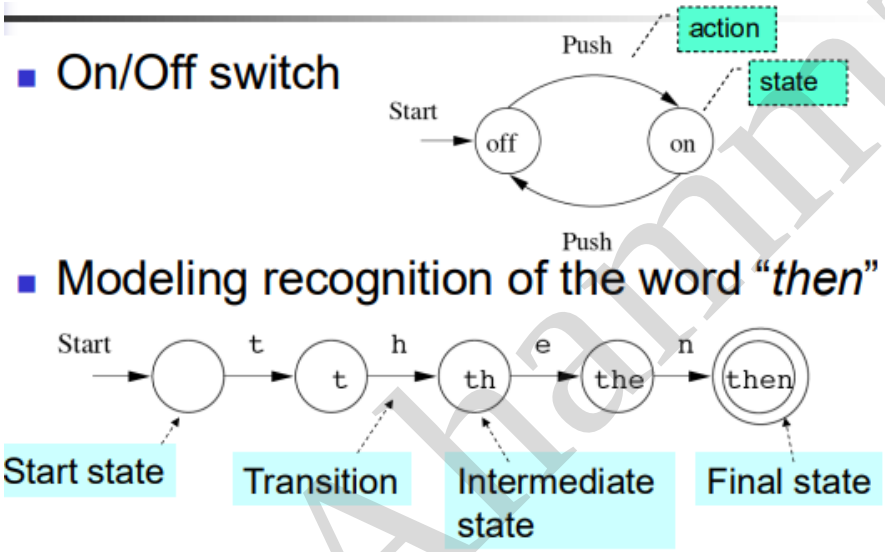
Yes, the string w=100011s **does belong** to the language of strings with an equal number of 0s and 1s because it has **3 zeros** and **3 ones**, which are equal.

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❖ **Applications of Finite Automata**

✔ **Digital Circuit Design** – Helps design and verify circuit behavior.
✔ **Lexical Analysis** – Used in compilers to process code.
✔ **Pattern Matching** – Scans large texts (e.g., web pages) for specific patterns.
✔ **System Verification** – Ensures correctness in stock transactions, network protocols, and other finite-state systems.

❖ **Example of finite automata:**

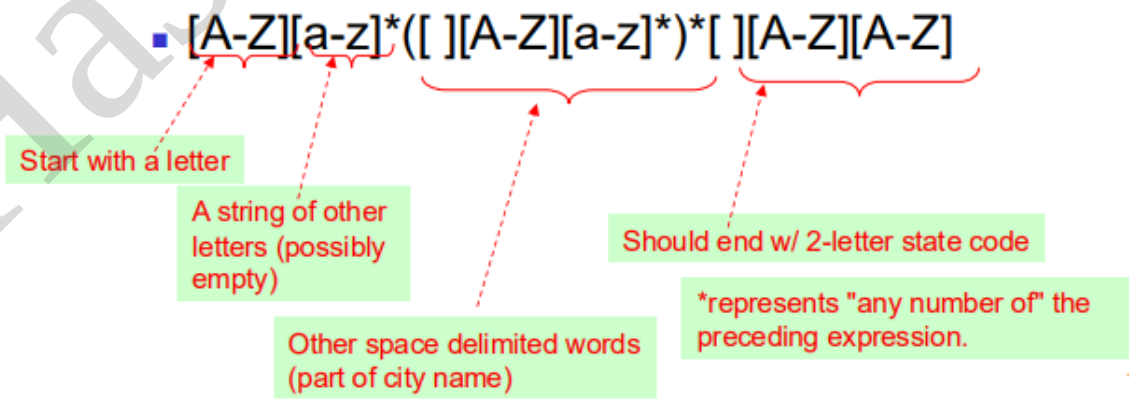


❖ Structural Expressions & Grammars

✔ Grammars are used in software that processes recursive structures.
✔ Parsers in compilers use grammars to handle nested expressions.
✔ **Example Rule:** E → E + E means an expression can be formed by adding two expressions.

❖ **Regular expressions**

➢ Denote the structure of data, especially text strings.
➢ E.g., unix style to capture city names such as **"Palo Alto CA"**

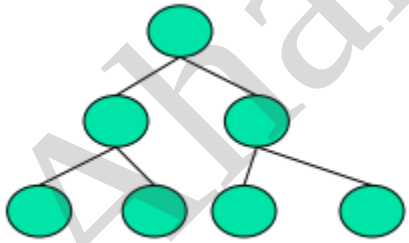


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- **Deductive proof:** Let,
  - **Claim-1:** If $y \geq 4$, then $2^y \geq y^2$. Let x be any number which is obtained by adding the squares of 4 positive integers.
  - **Claim-2**: Given x and assuming that Claim 1 is true, prove that $2^x \geq x^2$
    - Proof:
      1) Given: $x = a^2 + b^2 + c^2 + d^2$
      2) Given: $a \geq 1, b \geq 1, c \geq 1, d \geq 1$
      3) → $a^2 \geq 1, b^2 \geq 1, c^2 \geq 1, d^2 \geq 1$ (by 2)
      4) → $x \geq 4$ (by 1 & 3)
      5) → $2^x \geq x^2$ (by 4 and Claim 1)

    "implies" or "follows"

- **Theorems, Lemmas and Corollaries:**
  - **Theorem:** A major result as a "theorem"
    Example: the height of an n-node binary tree is at least floor(n)
  - **Lemma:** An intermediate result that we show to prove a larger result as a "lemma"
    Example: Level I of a perfect binary tree has 2^i nodes.
  - **Corollary:** A result that follows from an already proven result as a "corollary".
    Example: A perfect binary tree of height h has $2^{h+1} - 1$ nodes.



- **Quantifiers:** A quantifier is a word that usually goes before a noun to express the quantity of the object.
  - Universal quantifier: "For all" or "For every" and is denoted by- ∀
  - Existential quantifier: "There exists" and is denoted by- ∃.

    E.g., "IF A THEN B" can also be written as" A=>B
- **Proving techniques:**
  1. By contradiction
  2. By induction
  3. By contrapositive statement.
  4. By counter example
- **There is no such thing called a "proof by example"!**
  An example only shows one case, but a proof must apply to all cases.
  Example: Claim: "All prime numbers are odd."
  If you show that 3 and 5 are prime and odd, that's just an example. But 2 is a prime number and even, so the claim is false.

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❖ **Theorem-1.5: Let S be a finite subset of some infinite set U. Let T be the complement of S with respect to U. Then T is infinite.**

Let consider two definition for proving the theorem:

1. A set S is finite if it has exactly n elements for some integer n, written as ||S||=n. If a set is not finite, it is called infinite, meaning it has more elements than any integer can count.
2. If S and T are subsets of U, then T is the complement of S if:
   - S∪T=U (together, they cover all of elements)
   - S∩T=∅ (they have no common elements).

   Simply, T contains everything in U that is not in S.

PROOF: Intuitively, this theorem says that if you have an infinite supply of something (U), and you take a finite amount away (S), then you still have an infinite amount left. Let us begin by restating the facts of the theorem as in Fig. 1.4.

Original Statement	New Statement
S is finite	There is a integer n such that $\|S\| = n$
U is infinite	For no integer p is $\|U\| = p$
T is the complement of S	$S \cup T = U$ and $S \cap T = \emptyset$

Figure 1.4: Restating the givens of Theorem 1.5

We know that S and T are **disjoint** and their union gives the entire set U, so:

$$||S||+||T||=||U||$$

Since S is **finite**, let's say it has n elements:

$$||S||=n$$

Since U is **infinite**, there is no integer p such that:

$$||U||=p$$

Now, assume T is **finite** with m elements:

$$||T||=m$$

Then,

$$||U||=||S||+||T||=n+m$$

Since n+m is an integer, this contradicts the fact that ||U|| cannot be an integer. Thus, our assumption that T is finite must be **false**, meaning T is **infinite**.

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❖ **Proof by induction- Prove that $2^x \geq x^2$ for all $x \geq 4$ using mathematical induction.**

1. **Base Case** ($x = 4$):
   $2^4 = 16$ and $4^2 = 16$.
   Thus, $2^4 \geq 4^2$. ✅

2. **Inductive Hypothesis:**
   Assume $2^x \geq x^2$ holds for some $x \geq 4$.

3. **Inductive Step:**
   Prove $2^{x+1} \geq (x+1)^2$.
   Rewrite $2^{x+1} = 2 \cdot 2^x$.
   Using the hypothesis ($2^x \geq x^2$):

$$2^{x+1} = 2 \cdot 2^x \geq 2 \cdot x^2.$$

   Now, prove $2 \cdot x^2 \geq (x+1)^2$:
   Expand $(x+1)^2 = x^2 + 2x + 1$.
   Clearly, $2x^2 \geq x^2 + 2x + 1$ when $x \geq 4$. ✅

4. **Additional Justification:**
   From inequality $x \geq 2 + \frac{1}{x}$, observe:
   - For $x \geq 4$, $\frac{1}{x} \leq \frac{1}{4}$, so $2 + \frac{1}{x} \leq 2.25$.
   - Hence, $x \geq 2 + \frac{1}{x}$ holds for $x \geq 4$.
     This supports the claim that $2x^2 \geq (x+1)^2$ and confirms the proof.

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